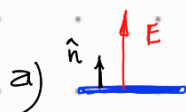
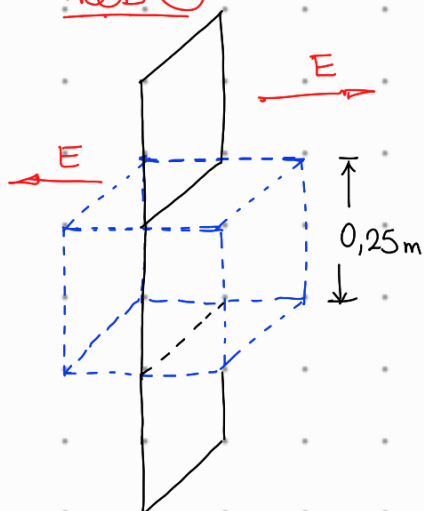


SOLUCIONARIO P.C N° 2

PROB ①



EL VECTOR $\hat{n}$ ES PERPENDICULAR A LAS CARAS DEL CUADRADO, ENTONCES EL FLUJO EN CADA UNA DE LAS CARAS ES:

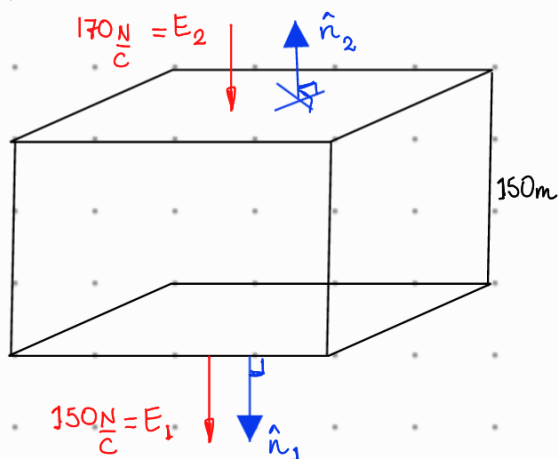
$$\Phi_E = \vec{E} \cdot \vec{S} = 100 \times (0,25)^2 = 6,25 \text{ V.m}$$

$$b) \quad \Phi_{\text{TOTAL}} = \frac{Q_{\text{ENCERRADA}}}{\epsilon_0}$$

$$2 \times 6,25 = \frac{Q_{\text{ENC}}}{\epsilon_0} \quad \Rightarrow \quad Q_{\text{ENC}} = 12,5 \epsilon_0 = 1,11 \times 10^{-10} \text{ C}$$

$$c) \quad \sigma = \frac{Q}{A} \quad \Rightarrow \quad \sigma = \frac{1,11 \times 10^{-10}}{0,25^2} \Rightarrow \sigma = 1,776 \times 10^{-9} \frac{\text{C}}{\text{m}^2}$$

PROB ②



$$a) \quad \Phi_1 = (150)(150)^2 \cos 0^\circ = 3375 \times 10^3 \text{ V.m}$$

$$\Phi_2 = (170)(150)^2 \cos 180^\circ = -3825 \times 10^3 \text{ V.m}$$

$$\Phi_{\text{TOTAL}} = -450 \times 10^3 \text{ V.m}$$

$$b) \quad \text{GAUSS:} \quad \Phi = \frac{Q}{\epsilon_0}$$

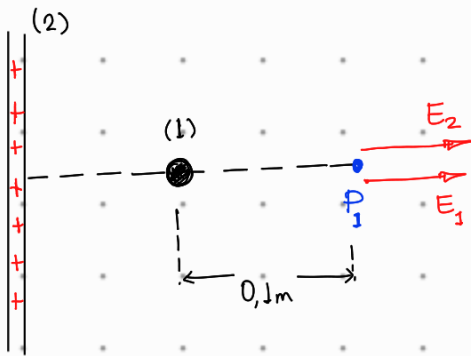
$$-450 \times 10^3 = \frac{Q}{\epsilon_0} \quad \Rightarrow \quad Q = -450 \times \epsilon_0 \times 10^3 = -3,98 \mu\text{C}$$

$$c) \quad \rho = \frac{Q}{V}$$

$$\rho = \frac{-3,98 \times 10^{-6}}{(150)^3}$$

$$\rho = -1,18 \times 10^{-12} \frac{\text{C}}{\text{m}^3}$$

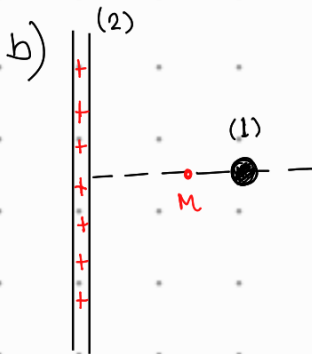
PROB (3)



$$a) E_1 = \frac{kQ_1}{|r|^2} = \frac{9 \times 10^9 \times 30 \times 10^{-9}}{(0,1)^2} \Rightarrow E_1 = 27 \times 10^3 \frac{N}{C}$$

$$E_2 = \frac{\sigma}{2\epsilon_0} = \frac{0,4 \times 10^{-6}}{2 \times (8,85 \times 10^{-12})} \Rightarrow E_2 = 22599 \frac{N}{C}$$

$$\vec{E}_{\text{Total}} = 49599 \frac{N}{C} \hat{c}$$

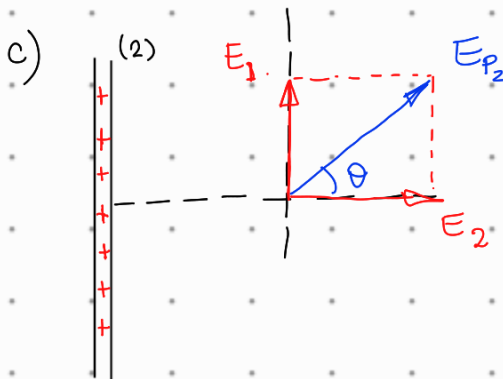


EN EL PUNTO "M":

$$E_1 = E_2$$

$$\frac{kQ}{r^2} = \frac{\sigma}{2\epsilon_0} \Rightarrow r = \sqrt{\frac{2\epsilon_0 kQ}{\sigma}}$$

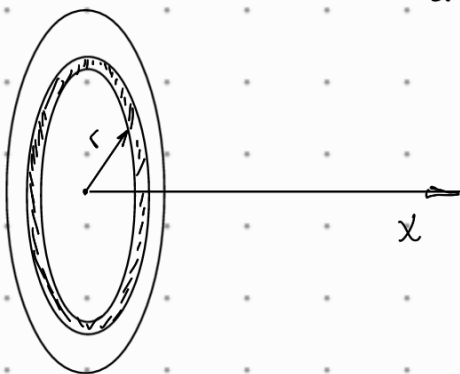
$$r = 0,11m$$



$$E_{P2} = 35209,58 \frac{N}{C}$$

$$\theta = \arctan\left(\frac{E_1}{E_2}\right) = 50^\circ$$

PROB (4)

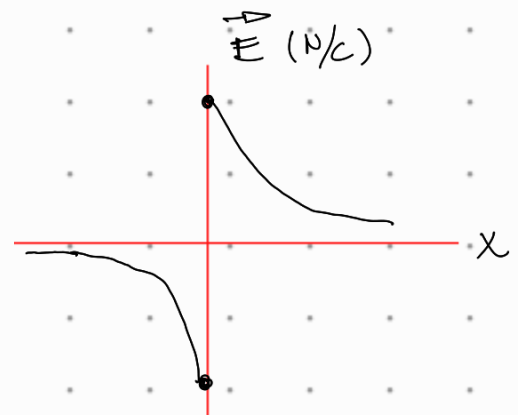


$$dE = \frac{k dq x}{(r^2 + x^2)^{3/2}}$$

$$dA = 2\pi r dr$$

$$dq = \sigma dA$$

$$\Rightarrow dq = 2\pi r \sigma dr$$



$$\Rightarrow E = 2\pi K \sigma x \int_0^R \frac{r dr}{(r^2 + x^2)^{3/2}} = 2\pi K \sigma \left[1 - \frac{x}{(R^2 + x^2)^{1/2}} \right]$$